

In []:

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# === Step 1: Load Data (fix header row) ===
file_path = r"C:\Users\Aditya\Videos\ASRS\motion control\pwm mapping\Code\Motor_1_Back.xlsx"
sheet_name = "Sheet1"

data = pd.read_excel(file_path, sheet_name=sheet_name, header=1) # row 2 = headers

# Check columns
print("Columns:", data.columns)

# === Step 2: Extract X and Y ===
x = data['RPM(Motor1_B)'].values # Input
y = data['Speed_percentage'].values # Output

# === Step 3: Try multiple polynomial fits ===
degrees = [3]

plt.figure(figsize=(8,6))
plt.scatter(x, y, color='black', label="Data points")

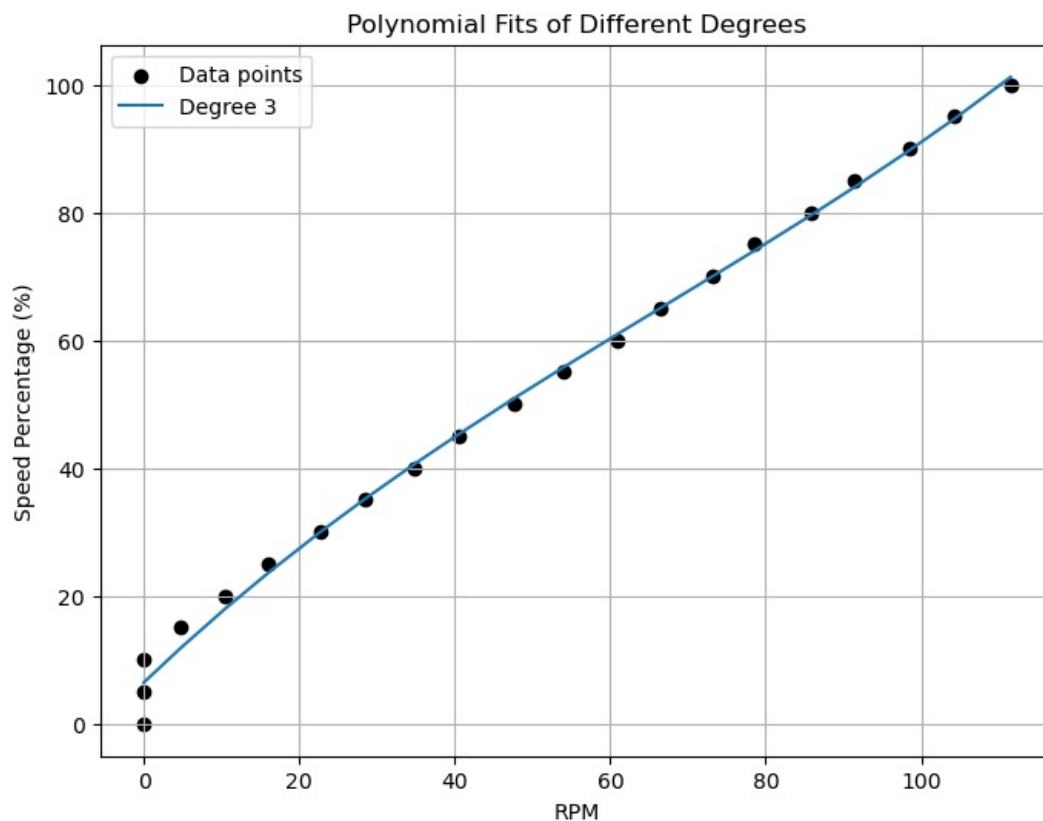
for d in degrees:
    coeffs = np.polyfit(x, y, d)
    poly = np.poly1d(coeffs)
    plt.plot(x, poly(x), label=f"Degree {d}")
    print(f"\nPolynomial (degree {d}): \n{poly}")

# === Step 4: Plot ===
plt.xlabel("RPM")
plt.ylabel("Speed Percentage (%)")
plt.title("Polynomial Fits of Different Degrees")
plt.legend()
plt.grid(True)
plt.show()
```

Columns: Index(['Unnamed: 0', 'RPM(Motor1_B)', 'Speed_percentage'], dtype='object')

Polynomial (degree 3):

$3.198 \times 10^{-5} x^3 - 0.006403 x^2 + 1.166 x + 6.434$



In [33]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# === Step 1: Load Data (fix header row) ===
file_path = r"C:\Users\Aditya\Videos\ASRS\motion control\pwm mapping\Code\Motor_1_Forward.xlsx"
sheet_name = "Sheet1"
```

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data = pd.read_excel(file_path, sheet_name=sheet_name, header=1) # row 2 = headers

# Check columns
print("Columns:", data.columns)

# === Step 2: Extract X and Y ===
x = data['RPM(Motor1_F)'].values # Input
y = data['Speed_percentage'].values # Output

# === Step 3: Try multiple polynomial fits ===
degrees = [3]

plt.figure(figsize=(8,6))
plt.scatter(x, y, color='black', label="Data points")

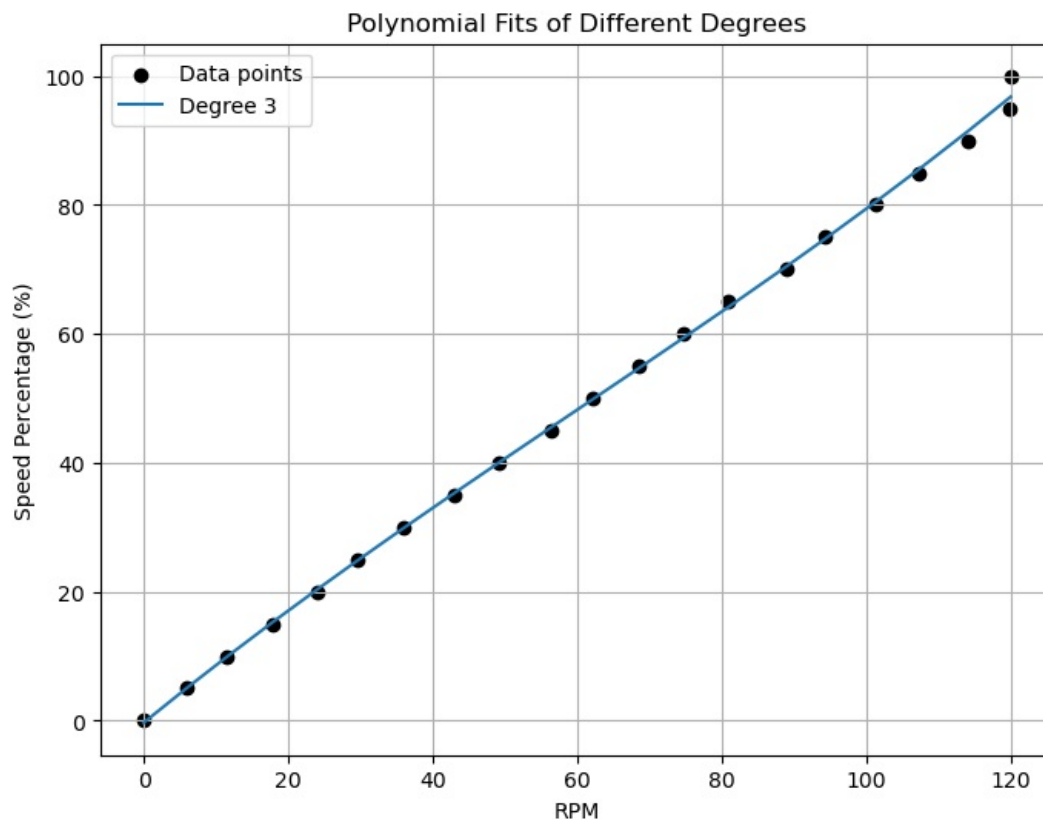
for d in degrees:
    coeffs = np.polyfit(x, y, d)
    poly = np.poly1d(coeffs)
    plt.plot(x, poly(x), label=f"Degree {d}")
    print(f"\nPolynomial (degree {d}): \n{poly}")

# === Step 4: Plot ===
plt.xlabel("RPM")
plt.ylabel("Speed Percentage (%)")
plt.title("Polynomial Fits of Different Degrees")
plt.legend()
plt.grid(True)
plt.show()

```

Columns: Index(['RPM(Motor1_F)', 'Speed_percentage'], dtype='object')

Polynomial (degree 3):

$$1.497\text{e-}05 x^3 - 0.002692 x^2 + 0.9166 x - 0.3084$$


```

In [40]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# === Step 1: Load Data ===
file_path = r"C:\Users\Aditya\Videos\ASRS\motion control\pwm mapping\Code\Motor_4_Forward.xlsx"
sheet_name = "Sheet1"

data = pd.read_excel(file_path, sheet_name=sheet_name, header=0) # row 1 = headers

print("Columns:", data.columns)

# === Step 2: Extract X and Y ===
x = data['RPM(Motor4_F)'].values
y = data['Speed_percentage'].values

# === Step 3: Polynomial fits ===
degrees = [3]

```

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plt.figure(figsize=(8,6))
plt.scatter(x, y, color='black', label="Data points")

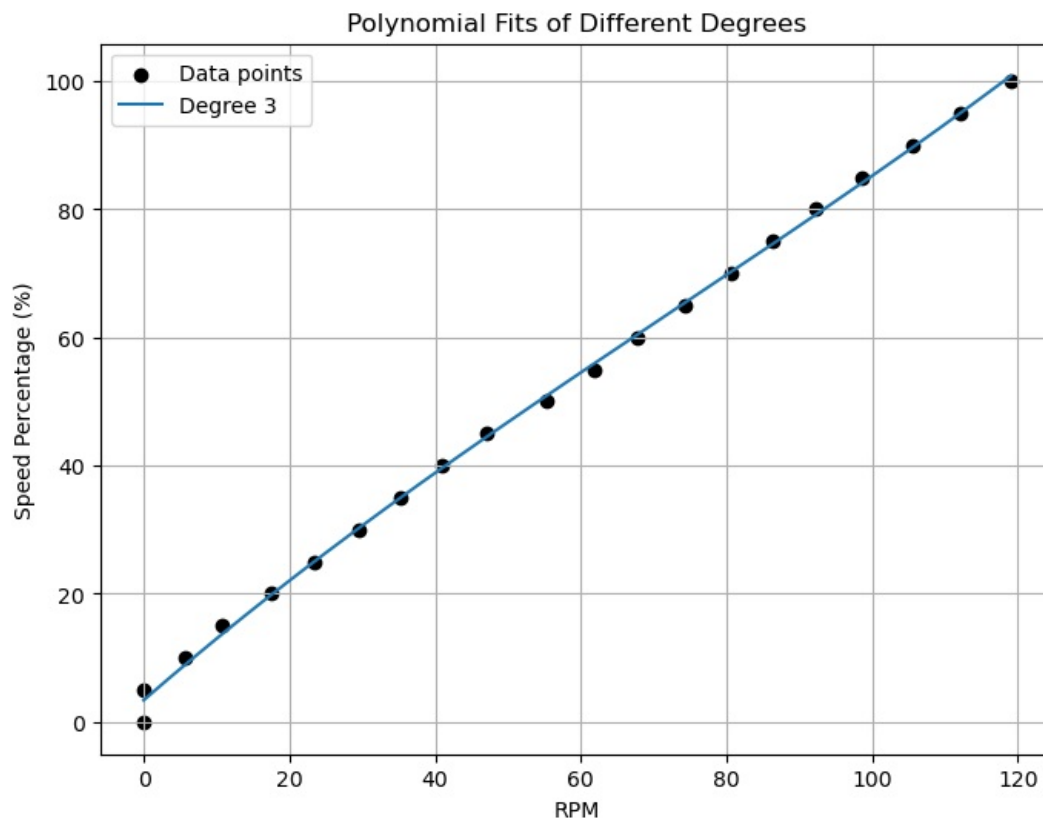
x_sorted = np.linspace(min(x), max(x), 500)

for d in degrees:
    coeffs = np.polyfit(x, y, d)
    poly = np.poly1d(coeffs)
    plt.plot(x_sorted, poly(x_sorted), label=f"Degree {d}")
    print(f"\nPolynomial (degree {d}): \n{poly}")

# === Step 4: Plot ===
plt.xlabel("RPM")
plt.ylabel("Speed Percentage (%)")
plt.title("Polynomial Fits of Different Degrees")
plt.legend()
plt.grid(True)
plt.show()
```

Columns: Index(['RPM(Motor4_F)', 'Speed_percentage'], dtype='object')

Polynomial (degree 3):

$$1.453e-05 x^3 - 0.003171 x^2 + 0.9904 x + 3.381$$


```
In [42]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# === Step 1: Load Data ===
file_path = r"C:\Users\Aditya\Videos\ASRS\motion control\pwm mapping\Code\Motor_4_Back.xlsx"
sheet_name = "Sheet1"

data = pd.read_excel(file_path, sheet_name=sheet_name, header=0) # row 1 = headers

print("Columns:", data.columns)

# === Step 2: Extract X and Y ===
x = data['RPM(Motor4_B)'].values
y = data['Speed_percentage'].values

# === Step 3: Polynomial fits ===
degrees = [3]

plt.figure(figsize=(8,6))
plt.scatter(x, y, color='black', label="Data points")

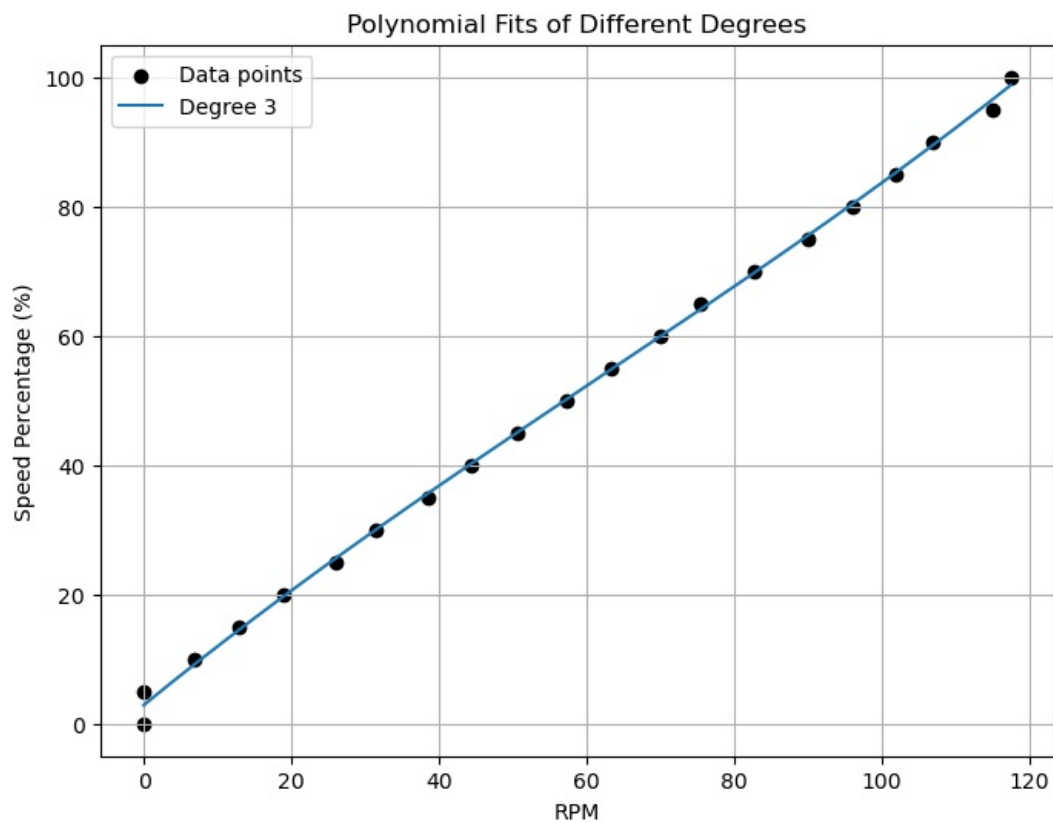
x_sorted = np.linspace(min(x), max(x), 500)

for d in degrees:
    coeffs = np.polyfit(x, y, d)
    poly = np.poly1d(coeffs)
    plt.plot(x_sorted, poly(x_sorted), label=f"Degree {d}")
    print(f"\nPolynomial (degree {d}): \n{poly}")
```

```
# === Step 4: Plot ===
plt.xlabel("RPM")
plt.ylabel("Speed Percentage (%)")
plt.title("Polynomial Fits of Different Degrees")
plt.legend()
plt.grid(True)
plt.show()

Columns: Index(['RPM(Motor4_B)', 'Speed_percentage'], dtype='object')
```

```
Polynomial (degree 3):
      3      2
1.485e-05 x - 0.002752 x + 0.9342 x + 2.871
```



```
In [44]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# === Step 1: Load Data ===
file_path = r"C:\Users\Aditya\Videos\ASRS\motion control\pwm mapping\Code\Motor_2_Back.xlsx"
sheet_name = "Sheet1"

data = pd.read_excel(file_path, sheet_name=sheet_name, header=0) # row 1 = headers

print("Columns:", data.columns)

# === Step 2: Extract X and Y ===
x = data['RPM(Motor2_B)'].values
y = data['Speed_percentage'].values

# === Step 3: Polynomial fits ===
degrees = [3]

plt.figure(figsize=(8,6))
plt.scatter(x, y, color='black', label="Data points")

x_sorted = np.linspace(min(x), max(x), 500)

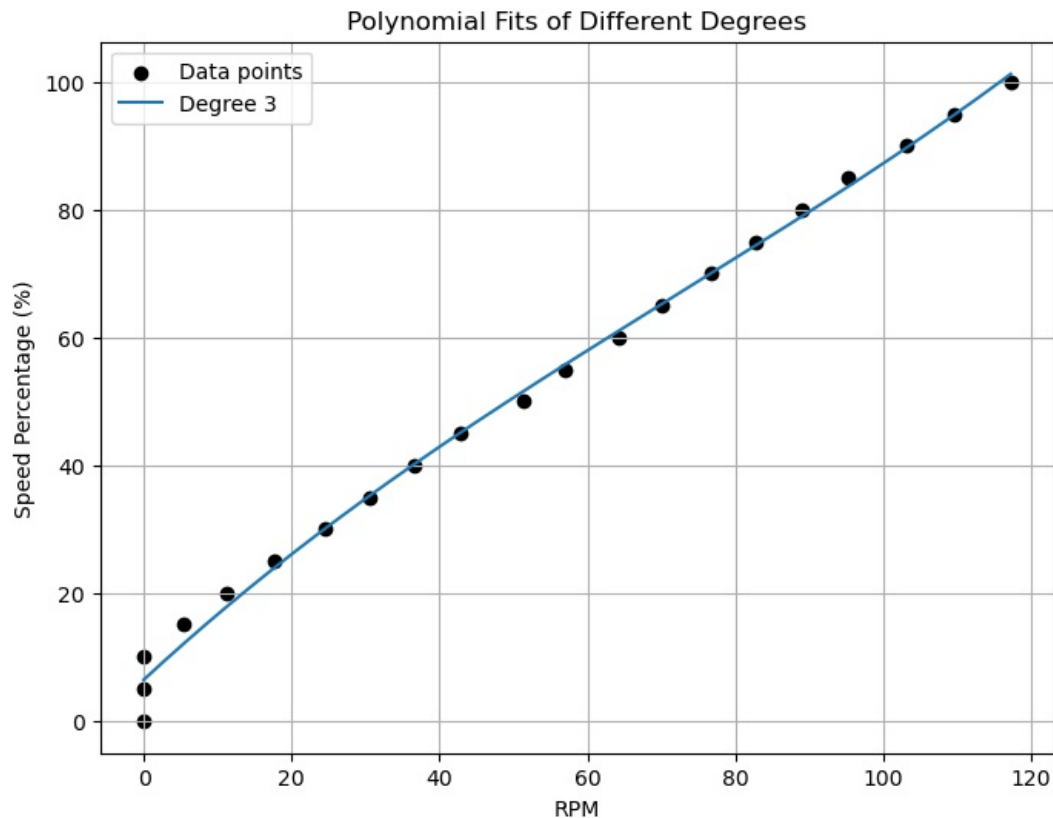
for d in degrees:
    coeffs = np.polyfit(x, y, d)
    poly = np.poly1d(coeffs)
    plt.plot(x_sorted, poly(x_sorted), label=f"Degree {d}")
    print(f"\nPolynomial (degree {d}): \n{poly}")

# === Step 4: Plot ===
plt.xlabel("RPM")
plt.ylabel("Speed Percentage (%)")
plt.title("Polynomial Fits of Different Degrees")
plt.legend()
plt.grid(True)
plt.show()
```

Columns: Index(['RPM(Motor2_B)', 'Speed_percentage'], dtype='object')

Polynomial (degree 3):

$$2.272e-05 x^3 - 0.004907 x^2 + 1.072 x + 6.4$$



```
In [46]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# === Step 1: Load Data ===
file_path = r"C:\Users\Aditya\Videos\ASRS\motion control\pwm mapping\Code\Motor_2_Forward.xlsx"
sheet_name = "Sheet1"

data = pd.read_excel(file_path, sheet_name=sheet_name, header=0) # row 1 = headers

print("Columns:", data.columns)

# === Step 2: Extract X and Y ===
x = data['RPM(Motor2_F)'].values
y = data['Speed_percentage'].values

# === Step 3: Polynomial fits ===
degrees = [3]

plt.figure(figsize=(8,6))
plt.scatter(x, y, color='black', label="Data points")

x_sorted = np.linspace(min(x), max(x), 500)

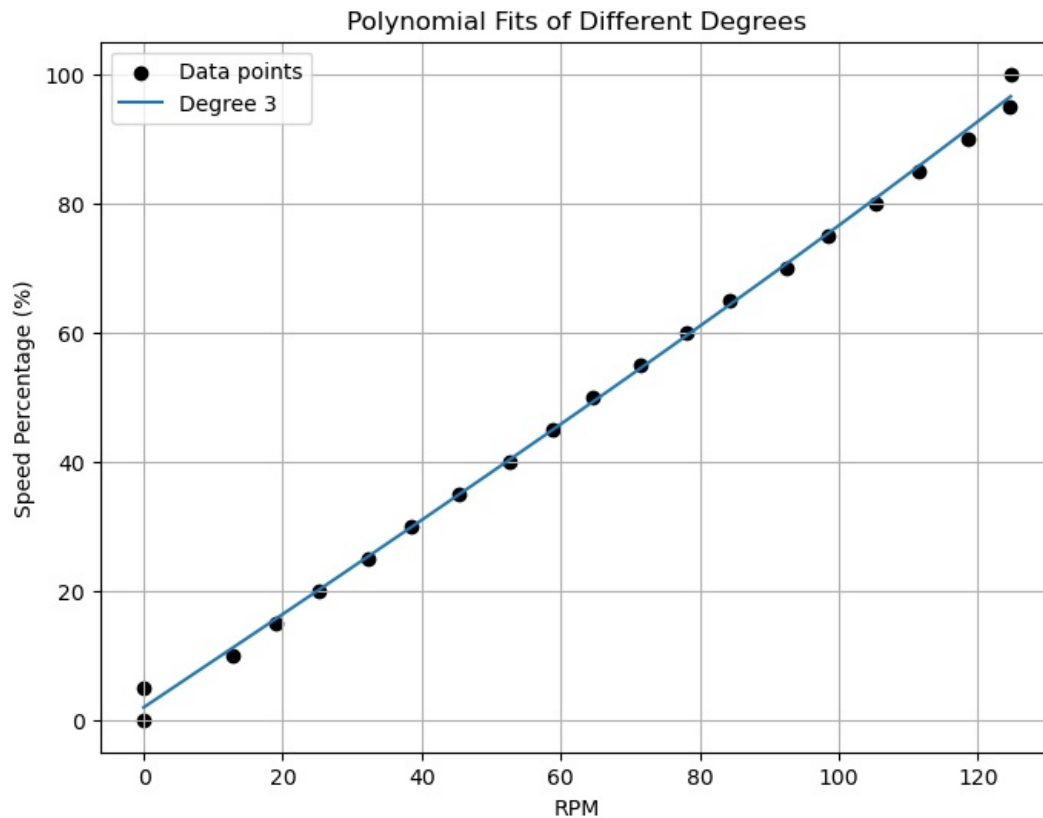
for d in degrees:
    coeffs = np.polyfit(x, y, d)
    poly = np.poly1d(coeffs)
    plt.plot(x_sorted, poly(x_sorted), label=f"Degree {d}")
    print(f"\nPolynomial (degree {d}): \n{poly}")

# === Step 4: Plot ===
plt.xlabel("RPM")
plt.ylabel("Speed Percentage (%)")
plt.title("Polynomial Fits of Different Degrees")
plt.legend()
plt.grid(True)
plt.show()
```

Columns: Index(['RPM(Motor2_F)', 'Speed_percentage'], dtype='object')

Polynomial (degree 3):

$$1.777e-06 x^3 + 0.0001007 x^2 + 0.7185 x + 1.926$$



```
In [48]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# === Step 1: Load Data ===
file_path = r"C:\Users\Aditya\Videos\ASRS\motion control\pwm mapping\Code\Motor_3_Forward.xlsx"
sheet_name = "Sheet1"

data = pd.read_excel(file_path, sheet_name=sheet_name, header=0) # row 1 = headers
print("Columns:", data.columns)

# === Step 2: Extract X and Y ===
x = data['RPM(Motor3_F)'].values
y = data['Speed_percentage'].values

# === Step 3: Polynomial fits ===
degrees = [3]

plt.figure(figsize=(8,6))
plt.scatter(x, y, color='black', label="Data points")

x_sorted = np.linspace(min(x), max(x), 500)

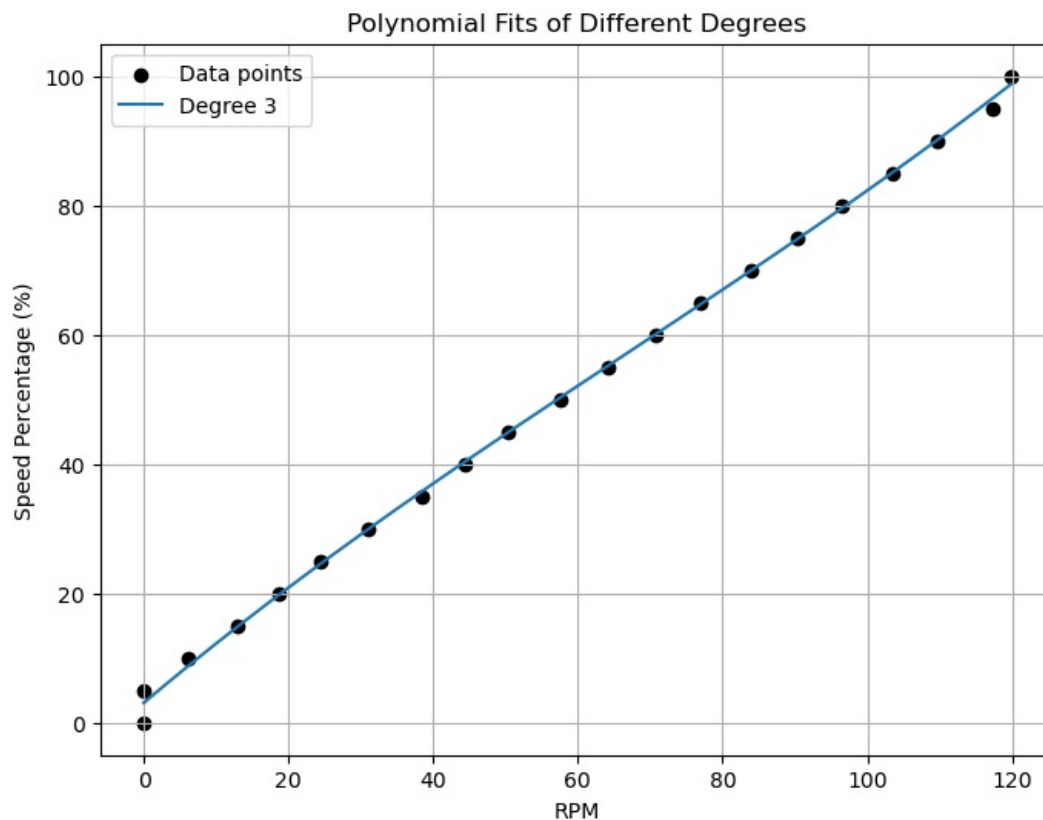
for d in degrees:
    coeffs = np.polyfit(x, y, d)
    poly = np.poly1d(coeffs)
    plt.plot(x_sorted, poly(x_sorted), label=f"Degree {d}")
    print(f"\nPolynomial (degree {d}): \n{poly}")

# === Step 4: Plot ===
plt.xlabel("RPM")
plt.ylabel("Speed Percentage (%)")
plt.title("Polynomial Fits of Different Degrees")
plt.legend()
plt.grid(True)
plt.show()
```

Columns: Index(['RPM(Motor3_F)', 'Speed_percentage'], dtype='object')

Polynomial (degree 3):

$$1.528e-05 x^3 - 0.003066 x^2 + 0.9464 x + 3.058$$



```
In [53]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# === Step 1: Load Data ===
file_path = r"C:\Users\Aditya\Videos\ASRS\motion control\pwm mapping\Code\Motor_3_Back.xlsx"
sheet_name = "Sheet1"

data = pd.read_excel(file_path, sheet_name=sheet_name, header=0) # row 1 = headers

print("Columns:", data.columns)

# === Step 2: Extract X and Y ===
x = data['RPM(Motor3_B)'].values
y = data['Speed_percentage'].values

# === Step 3: Polynomial fits ===
degrees = [3]

plt.figure(figsize=(8,6))
plt.scatter(x, y, color='black', label="Data points")

x_sorted = np.linspace(min(x), max(x), 500)

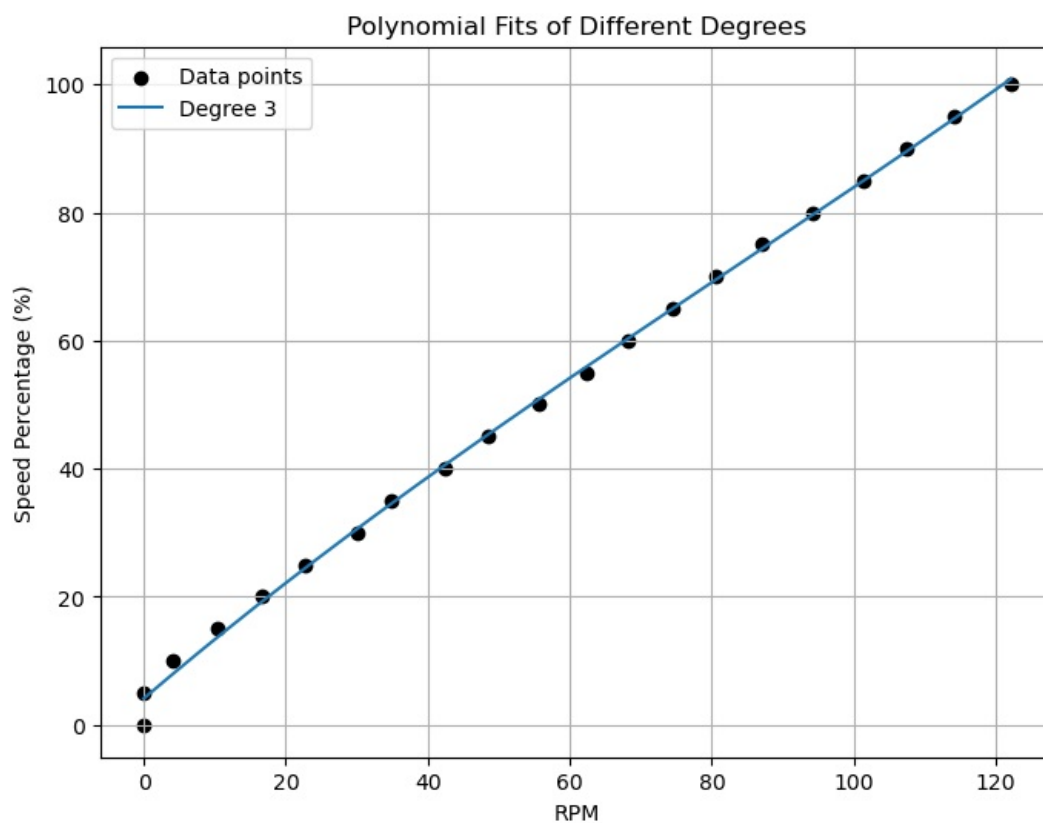
for d in degrees:
    coeffs = np.polyfit(x, y, d)
    poly = np.poly1d(coeffs)
    plt.plot(x_sorted, poly(x_sorted), label=f"Degree {d}")
    print(f"\nPolynomial (degree {d}): \n{poly}")

# === Step 4: Plot ===
plt.xlabel("RPM")
plt.ylabel("Speed Percentage (%)")
plt.title("Polynomial Fits of Different Degrees")
plt.legend()
plt.grid(True)
plt.show()
```

Columns: Index(['RPM(Motor3_B)', 'Speed_percentage'], dtype='object')

Polynomial (degree 3):

$$9.995e-06 x^3 - 0.002492 x^2 + 0.9476 x + 4.095$$



In []:

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